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RESEARCH ARTICLE

Auditory stimulation during deep sleep enhances total slow-wave activity in a young cohort: A feasibility trial

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Summary

Cognitive, metabolic and sleep benefits associated with enhancement of sleep slow waves using closed-loop auditory stimulation have been reported in adults but not in adolescents, especially in home settings. Seventeen volunteers (10F/7M; age range: 13–18 years old) participated in a 2-week, single-blind, crossover study. STIM (auditory stimulation ON) and SHAM (auditory stimulation at zero-volume) were each applied for a week (randomized order). Participants used a self-applied, single-electroencephalogram, wearable device at home. An embedded algorithm performed real-time sleep staging, detected slow-wave sleep and delivered auditory tones separated by a 1-s inter-tone interval. After each sleep session, participants completed questionnaires to report sleep quality, sleepiness (Karolinska Sleepiness Scale), and performed tasks to quantify vigilance (Psychomotor Vigilance Task) and working memory (continuous working memory performance task). Sleep architecture, count of microarousals, slow-wave amplitude, total and mean slow-wave activity (electroencephalogram power in the 0.5–4 Hz frequency band) during non-rapid eye movement sleep, sleepiness level, and cognitive performance metrics were compared between STIM and SHAM. The slow-wave amplitude during stimulation, total slow-wave activity and mean slow-wave activity were significantly higher in the STIM condition (+10.7%, +7.38% and + 7.57%). The count of microarousals, and the power in alpha and beta bands were not different between SHAM and STIM. The Pearson correlation between slow-wave activity enhancement and sleep duration (−0.83; $p < 1e - 4$) suggested a significant decrease in sleep duration proportional to slow-wave activity enhancement. Trending results ($p < 0.1$) in the STIM condition included higher number of correct continuous working memory performance task responses (+1.01 correct; $p = 0.07$). This research provides feasibility of auditory stimulation-based slow-wave activity in a young population.

KEYWORDS
auditory, enhancement, slow-wave sleep, stimulation, young cohort

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1 | INTRODUCTION

During adolescence there is a stable sleep need of approximately 10 hr per night. However, this critical developmental period is confronting a pervasive issue of sleep deprivation (Owens & Weiss, 2017; Paruthi et al., 2016; Watson et al., 2015). The habitual restriction of sleep duration, particularly on school nights, has given rise to a concerning public health epidemic (Carskadon, 2011). This trend has been underscored by the American Academy of Sleep Medicine (AASM), recognizing insufficient adolescent sleep as a serious health concern (Paruthi et al., 2016).

Insufficient sleep during adolescence has deleterious effects on daytime cognitive (Lo et al., 2016; Lo & Chee, 2020; Short & Chee, 2019) and affective (Palmer & Alfano, 2017) function, and is implicated in the development, progression and maintenance of mood and anxiety disorders (Gradisar et al., 2022; Jamieson et al., 2021; Uccella et al., 2023). The intricate relationship between sleep problems and mental health underscores the critical need for interventions that enhance sleep quality among adolescents.

During a night of sleep, the brain cycles between alternating periods of rapid eye movement (REM) sleep and non-rapid eye movement (NREM) sleep, with NREM sleep occupying approximately 70% of total sleep time (TST) in adolescents.

A wealth of evidence suggests that the restorative work of sleep occurs during the deepest stage of NREM sleep where slow waves are more abundant (Bellesi et al., 2014). The amplitude and number of slow waves can be quantified by the slow-wave activity (SWA), which represents the electroencephalogram (EEG) power in the delta frequency band (0.5–4 Hz). Slow waves are hallmarks of NREM, and their spatial distribution, temporal density and amplitude are positively associated with learning, memory and the restorative value of sleep (Akerstedt et al., 1997; Dijk, 2009; Huber et al., 2004; Tononi & Cirelli, 2006, 2014; Figure 1).

Slow-wave activity has been associated with the homeostatic component "S" of the two-process model of sleep regulation, which considers a circadian "C" component as well (Achermann & Borbély, 2003). According to this model, sleep need is higher at the beginning of a sleep session and progressively decreases as sleep continues (Figure 2).

The amplitude and temporal density of sleep slow waves can be enhanced through auditory stimulation (Bellesi et al., 2014). An increase in SWA due to auditory stimulation was shown to be significantly correlated with enhanced memory consolidation in adults aged 23–42 years (Leminen et al., 2017), and memory improvement in adults over 60 years. (Papalambros et al., 2017). Consecutive nights of acoustic stimulation enhanced SWA, and improved next-day attention and alertness in a cohort of 51 chronically short sleepers (age range: 22–51 years; Diep et al., 2021).

This article reports the results of a study (SLEEPY) that was conducted to evaluate the feasibility of applying closed-loop auditory stimulation to enhance SWA in a cohort of sleep-restricted 13–18-year-old adolescents with daytime sleepiness. A secondary goal of the study was to evaluate the effect of SWA enhancement on cognitive function and sleep architecture.

2 | METHODS

2.1 | Study design

Youths were selected based on self-reported sleep restriction, characterized as: (1) consistently sleeping fewer than 8.0 hr on school nights; (2) engaging in weekend catch-up sleep by extending the morning sleep period by 2 or more hours when possible. The presence of longer sleep duration on days with extended sleep opportunity was confirmed by the guardian. In addition, all youths completed the Cleveland Adolescent Sleepiness Questionnaire (CASQ; Spilsbury et al., 2007), a validated 16-item instrument to measure daytime sleepiness in adolescents aged 11–17 years. The CASQ has two dimensions: Sleepiness Statements and Alertness Statements; items are scored from 1 (never) to almost every day (5 or more). To produce an overall score (items related to alertness were reversed), participants' responses are aggregated. The CASQ scores range between 16 and 80. Although there is no formal cut-off indicating sleepiness,

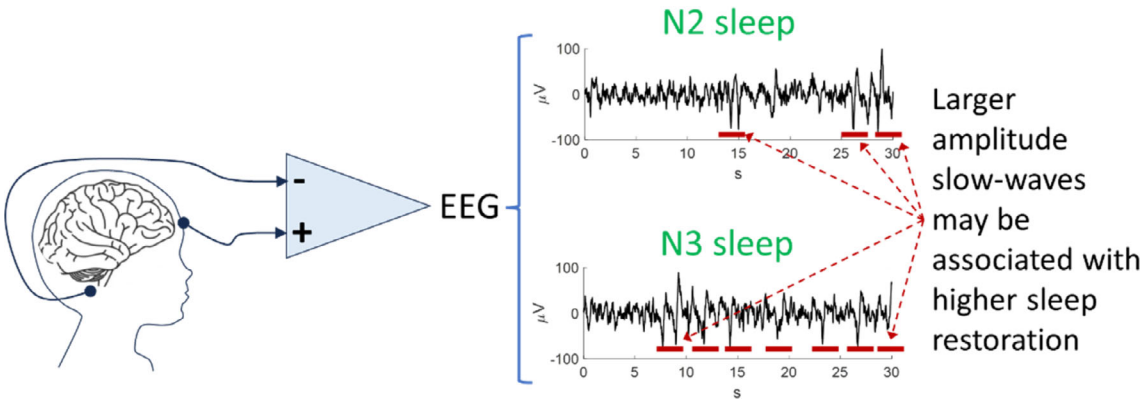

FIGURE 1 Sleep slow waves manifest in the electroencephalogram (EEG) during non-rapid eye movement (NREM) sleep, and are more abundant during N3 sleep.

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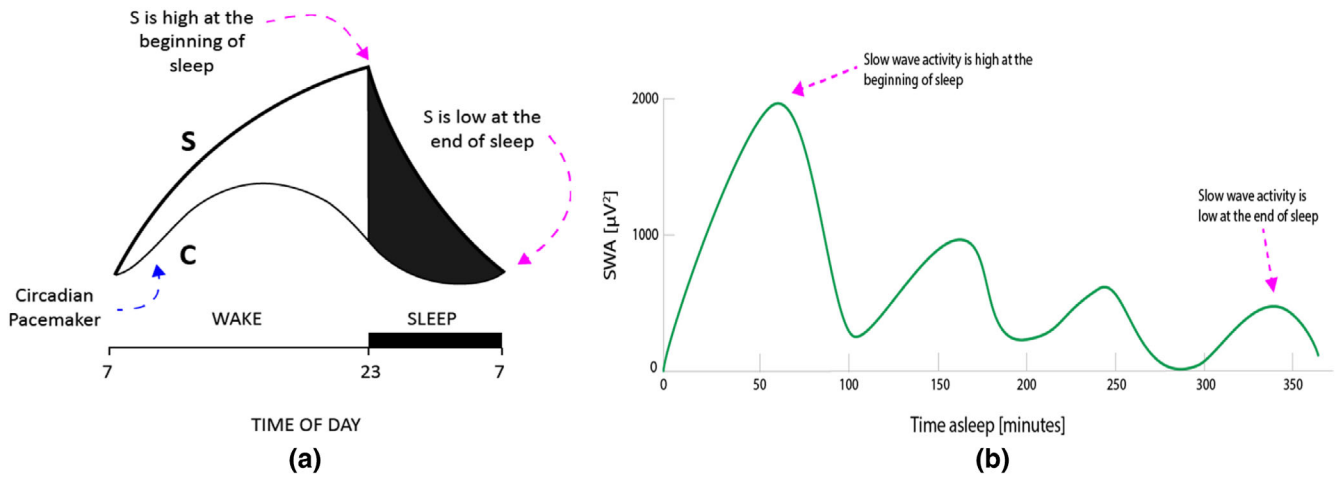
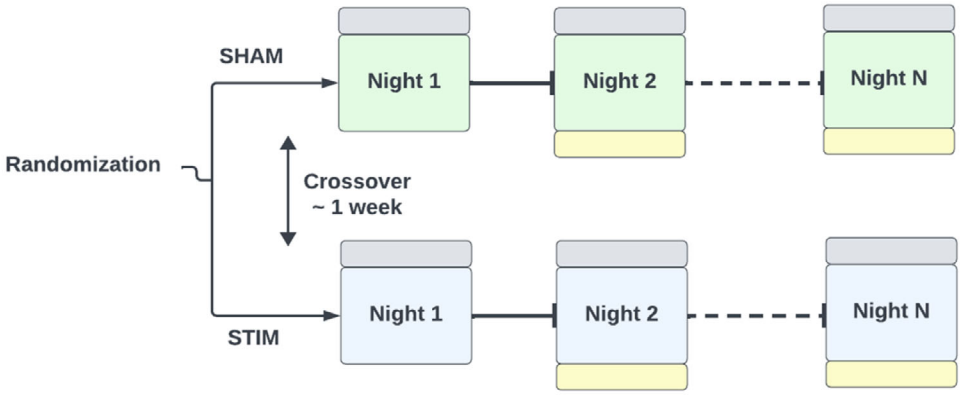


FIGURE 2 (a) Two-process model of sleep regulation. (b) Slow-wave activity (SWA) dynamics throughout a sleep session.

FIGURE 3 Subjects were randomly assigned to either a SHAM or STIM condition for week 1, followed by the other condition during week 2.



	STIM: night with auditory stimulation
	SHAM: identical to STIM nights except that the volume of stimulation is set to zero
	Evening sleep quality questionnaire, Karolinska sleepiness scale (KSS), vigilance (PVT), working memory (N-BACK)
	Morning sleep quality questionnaire, Karolinska sleepiness scale (KSS), vigilance (PVT), working memory (N-BACK)

higher scores (> 30) indicate greater daytime sleepiness (Spilsbury et al., 2007).

Seventeen volunteers, 10 girls and 7 boys, mean age 16.1 years (SD: 1.8), were recruited for this study. The study was approved by the University of Wisconsin-Madison Institutional Review Board (#2015–0337). The mean CASQ value for this group was 34.6 (SD: 3.6).

An integrated closed-loop device (referred to as Smart Sleep) was developed by Philips and the University of Wisconsin-Madison to monitor sleep using a single EEG lead acquired at the frontal location (FpZ) referenced to the right mastoid (M2). The device is AC coupled with two cascaded single pole filters with high-pass cutoff frequencies of 0.16 Hz and 0.17 Hz, respectively. An adjustable headband holds the frontal electrode and two speakers on each side to present the auditory stimulation. A short cable stemming from the right side of the headband is used to connect the reference electrode to the right mastoid. Processing of the signals takes place in an embedded microprocessor.

The participants used the device at home for a period of 2 weeks with counterbalanced sham (SHAM) or stimulation (STIM) conditions, which were applied for 5 nights each week (Figure 3). On average, the number of days separating the two conditions was 3.4.

Participants were instructed to wear the headband and turn the device ON at bedtime by pressing a power button located on the front of the device. A brief auditory tone rendered through Smart Sleep's speakers confirmed the ON state. Participants turned OFF the device after waking up by pressing the power button for a few seconds until a tone was heard, indicating that the device transitioned into the OFF state.

The order of conditions was counterbalanced with subjects randomly assigned to either a SHAM or STIM condition for week 1.

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The SHAM condition differed from the STIM condition only in the fact that the volume of the stimulation in the former was set to 0. However, for analysis purposes, the volumes at which the stimulation was delivered in STIM and would have been delivered in SHAM were stored.

Additional outcomes were electronically collected using a hand-held device each day before and following study nights. The detailed analysis of these is the focus of a follow-up article. This paper focuses on the analysis of the EEG data, sleep architecture and sleepiness, as measured by the Karolinska Sleepiness Scale (KSS; Kaida et al., 2006); cognitive domains quantified by the Psychomotor Vigilance Task (PVT; Dinges & Powell, 1985); and the continuous working memory performance task (N-BACK; Kirchner, 1958).

A subset of eight participants consented to spend a single night in the sleep lab to collect a standard polysomnography (PSG) recording while simultaneously wearing the Smart Sleep device in SHAM mode. The purpose of acquiring PSG signals was to test the ability of Smart Sleep to detect sleep stages in real time in the young population that participated in this study.

2.2 | Closed-loop processing to enhance slow-wave sleep

Smart Sleep's EEG signal was acquired at 1000 Hz, notch filtered at 60 Hz to remove the powerline interference, band-pass filtered at 0.3–40 Hz, and downsampled to 100 Hz for real-time processing. To detect sleep stages, 6-s windows were processed by an embedded deep-neural network (DNN) that detected four stages: wake; REM sleep; light NREM; and deep NREM. The DNN consisted of a convolutional group composed of three layers and a recurrent group composed of three long-short-term-memory layers (Figure 4a). The specific details of the real-time algorithm for sleep staging on the basis of 6-s windows are described in Bresch et al. (2018) and Garcia-molina et al. (2019).

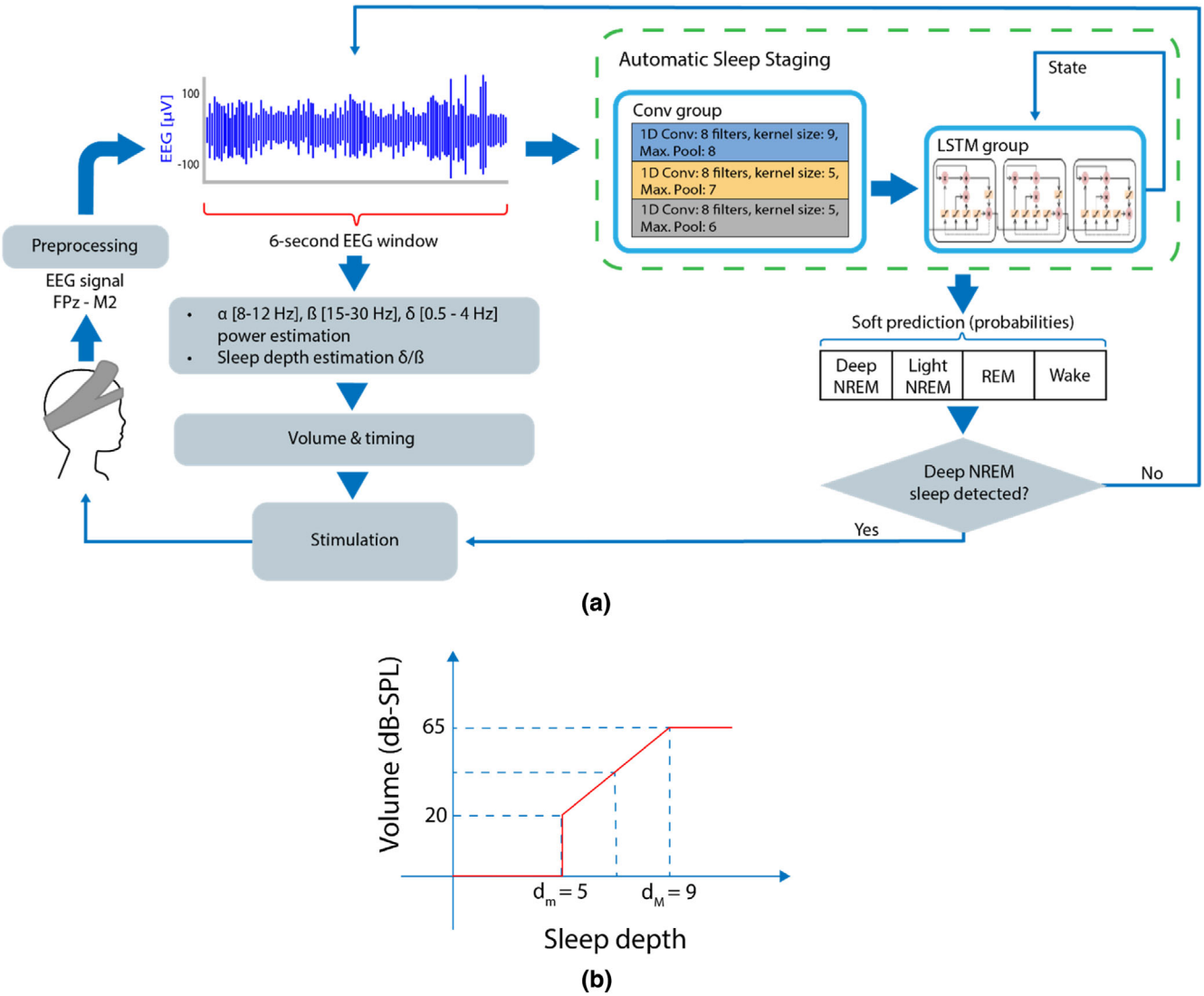

FIGURE 4 Closed-loop system (Smart Sleep) to enhance slow-wave sleep. (a) Data processing flow. (b) Volume in dB-SPL units versus sleep-depth.

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The accuracy of the DNN for sleep staging was verified using data from 22 participants (mean age 37.2 years; SD: 6.2 years) with similar demographic characteristics to the data used to design the algorithm (Garcia-Molina et al., 2019) compared with the manual annotation of two experts. A mean Kappa value of 0.67 resulted, which was close to the Kappa value of 0.69 generated between the two experts. The mean sensitivity and specificity for the detection of deep sleep were 0.78 and 0.95, respectively. The latter being as high as 0.95 is especially important to prevent stimulation from being delivered outside periods of deep sleep.

A parallel branch of the EEG processing filtered the signal in three frequency bands (alpha [8–12 Hz], beta [15–30 Hz] and delta [0.5–4 Hz]) using 2nd-order Chebyshev filters. The resulting signals were squared, then averaged over 1-s-long temporal windows for alpha and beta, and 10-s-long temporal windows for delta. The square root was then calculated to obtain the root mean square (RMS) power in each band. To detect sleep microarousals, the alpha and beta RMS values were compared with upper thresholds (10 μV and 3 μV, respectively), as described in Garcia-Molina et al. (2018). An estimation of sleep depth was obtained using the log-ratio between delta and beta RMS power.

If deep-NREM sleep was detected and the alpha and beta bands were below the microarousal detection threshold, auditory stimulation was delivered in the form of 50-ms tones separated from each other by a fixed 1-s interval. The pitch of each tone was randomized using a uniform distribution across the integer numbers in the frequency range from 500 to 2000 Hz to prevent habituation (Bellesi et al., 2014). The volume of each tone was determined depending on sleep-depth, as described in Garcia-Molina et al. (2018).

Figure 4(b) shows that louder tones were played during deeper sleep. The volume versus sleep-depth relationship was the same for all participants in the study. For sleep-depth values below the lower threshold ($d_m = 5$), no stimulation was delivered. For sleep depth values "d" between the lower threshold and the upper sleep-depth threshold ($d_M = 9$), the volume was set to $20 + 11.25 \times (d - 5)$ decibels of sound pressure (dB-SPL). For sleep-depth exceeding d_M, the volume was capped to 65 dB-SPL. The thresholds d_m and d_M were set considering the average sleep-depth dynamics reported in Garcia-Molina et al. (2016).

2.3 | Evaluation of the sleep staging algorithm on a young population

The PSG data collected using the Compumedics sleep system (Charlotte, NC, USA) along with the Smart Sleep data, on the eight participants who spent a night in the sleep lab, were used to evaluate the ability of the device to detect sleep stages in a young population. The PSG data were scored according to AASM (Berry et al., 2015) rules by an expert sleep technician.

The automatic sleep staging provided by Smart Sleep was manually aligned (via a consensus agreement among authors) with the technician-made sleep staging (which used the data from the Compumedics system) by considering multiple prominent EEG events such as movement-related signal deflections. Such events manifest in both the Compumedics and Smart Sleep signals. A linear interpolation stretch correction was also applied by aligning the beginning and end of recordings.

TABLE 1 Sleep stage mapping between Smart Sleep and manual staging.

Manually scored sleep stage	Smart Sleep corresponding stage
Wake	Wake
REM	REM
N1	Light NREM
N2	Light NREM
N3	Deep NREM

NREM, non-rapid eye movement; REM, rapid eye movement.

Given that Smart Sleep assigns sleep stages based on 6-s windows, the manual staging was up-sampled by a factor of 5 by simply repeating each sleep stage five times. This enabled us to compare the hypnogram generated by Smart Sleep and the manual up-sampled hypnogram. Mean sensitivity and specificity for each sleep stage and mean Kappa were calculated. Light NREM sleep detected by Smart Sleep corresponds to manually scored N1 or N2 stages. Thus, for comparison purposes, four stages were considered according to the mapping in Table 1. Bland–Altman analyses were also performed (Bland & Altman, 1986) to determine the limits-of-agreement for TST and deep NREM sleep. The device always produces a sleep stage estimation, even for windows containing movement artefacts (which are typically detected as Wake).

2.4 | Power spectrum analyses and SWA calculation

The EEG stored in the device was analysed (offline) using the 6-s-based DNN's sleep staging as reference. We refer to a 6-s segment to which a sleep stage was assigned as an epoch. For this analysis, NREM epochs (including deep and light sleep) were considered. The power spectrum density (PSD) per epoch was calculated using the Welch method (Welch, 1967) with a 4-s-long Hann window (ensuring 0.25-Hz frequency resolution), a 2-s-long overlap, and 512 points to calculate the Fourier transform.

The power in the alpha band was estimated using the epoch's PSD, and by integrating over the frequency range spanning from 8 to 12 Hz. The power values in the beta and sigma bands were also estimated from the PSD by integrating the frequency ranges spanning from 15 to 30 Hz and 11 to 11 Hz, respectively. The SWA of each epoch was estimated by integrating over the frequency range spanning from 0.5 to 4 Hz.

Detected sleep microarousals were considered to adjust the SWA calculation. Indeed, microarousal events have spike-like time–domain characteristics that manifest as high values overall in the spectral

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domain. Thus, 6-s segments containing detected-arousals were discarded from SWA analysis. The mean SWA was calculated by taking the average SWA over all considered NREM epochs, and the total SWA was calculated by adding SWA over all considered NREM epochs.

2.5 | Slow-wave count and amplitude

Individual slow waves in deep NREM sleep were detected using a similar algorithm to the one proposed by Ngo et al. (2013). In summary, events with a negative-going zero-crossing followed by a positive-going zero-crossing occurring between 800 and 1200 ms and with a minimum value lower than 80 microvolts qualified as slow waves.

The count and amplitude of detected slow waves during deep NREM sleep and during the periods of stimulation were compared between SHAM and STIM conditions. In the SHAM condition, the time during which stimulation would have been delivered was considered for the during-stimulation analysis.

2.6 | Sleep architecture analysis

Sleep architecture was analysed using the hypnogram generated by the DNN. Latency to light NREM, deep NREM and REM were determined from the moment where the device was turned on (ON) to the first epoch detected as light NREM, deep NREM and REM, respectively. Total sleep duration in minutes was estimated as 0.1 (i.e. 6 s) × the total number of epochs detected as sleep between ON and the time at which the device was turned off (OFF). The duration of a given detected stage in minutes was estimated as 0.1 × the total number of epochs detected as the targeted stage between the ON and OFF times. The wake after sleep onset (WASO) duration in minutes was estimated as 0.1 × the total number of epochs detected as Wake between the first and last epoch detected as sleep.

Because auditory stimulation was applied in deep NREM sleep, the analysis of the stability of this sleep stage was performed by quantifying the mean duration of bouts of deep NREM sleep. A bout is a sequence of epochs detected as deep NREM sleep; thus, the epoch before the start and after the end of a bout is of a sleep stage other than deep NREM.

The number of wake segments that were at least 3 min in duration in between the first and last epoch detected as sleep were also quantified to analyse a potential sleep disruption due to stimulation. In addition, the count of detected microarousals in NREM and REM for SHAM and STIM were compared to assess whether the stimulation has a sleep-disruptive effect.

2.7 | PVT, N-BACK and KSS analysis

The mean reaction time (RT) for a PVT session was calculated as the average in milliseconds of all RTs for valid individual trials in the session. A trial was considered valid (as recommended in Basner et al., 2011) if its RT was ≥ 100 ms (shorter RTs were considered false starts) and < 500 ms (longer RTs were considered as lapses). The number of lapses and false starts were also considered for further analysis.

To determine the mean value of the PVT- RT in SHAM or STIM conditions for each study participant, the mean RTs for all PVT sessions from that participant in SHAM or STIM conditions, respectively, were calculated. The PVT-RTs for SHAM and STIM conditions were analysed at the group level. A similar aggregation process was followed for the number of lapses and false starts.

The number of correct responses in a N-BACK session was used as the performance metric for that session. The mean number of correct responses for a participant's SHAM and STIM sessions were used for the group-level analysis.

The mean KSS values for a participant's SHAM and STIM sessions were calculated and analysed at the group level.

2.8 | Correlation between mean SWA, total SWA and sleep architecture, sleepiness or cognitive outcomes

To identify associations between the effect of the stimulation on sleep architecture, sleepiness or cognitive outcomes, the Pearson's correlations were calculated between the percentage of STIM to SHAM change in amplitude of slow waves, mean SWA and total SWA versus metrics of sleep architecture, sleepiness and/or cognitive outcomes.

The change in a slow-wave metric "*E*" between SHAM and STIM conditions was quantified as shown in Equation (1).

$$E_{STIM/SHAM}[\%] = 100 \times \left(\frac{E_{STIM}}{E_{SHAM}} - 1\right), \quad (1)$$

where E is the mean SWA, total SWA or the amplitude of slow waves. E_{STIM} and E_{SHAM} correspond to the values of E during STIM and SHAM, respectively.

2.9 | Statistical methods

For each sleep architecture metric (Table 3), mean SWA, total SWA and cognitive domain (Table 4) SHAM and STIM values were statistically compared using the paired, non-parametric Wilcoxon test (Rey & Neuhäuser, 2011). A comparison was considered statistically significant if the corresponding p-value was < 0.05.

To assess the statistical significance of the associations between slow-wave enhancement effects and sleep architecture or cognitive outcomes, the t-statistic with $n - 2$ degrees of freedom (Cohen et al., 2003; Equation 2) was used.

$$t = r\sqrt{\frac{n-2}{1-r^2}}, \quad (2)$$

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where r is the Pearson correlation between the slow-wave enhancement metric and the outcome, and n (= 17) is the number of study participants.

3 | RESULTS

3.1 | Evaluation of the sleep staging algorithm provided by Smart Sleep on the young population

The mean Kappa value was 0.62, which is lower compared with that of the older population (0.66) reported in Garcia-Molina et al. (2019). The results of the sleep stage performance are summarized in Table 2. Compared with the results of the validation of the sleep staging on an adult population (Garcia-Molina et al., 2019), the sensitivity to detect deep NREM is higher (0.86 versus 0.78), even while maintaining a high deep NREM specificity (0.94 versus 0.94), thus preventing stimulation from being delivered in lighter sleep stages and avoiding sleep disruption. The sensitivity to detect Wake, however, is lower in the younger population (0.60 versus 0.73), and the specificity to detect Wake is similar between the younger and older populations (0.95 versus 0.93). The mean Kappa value was 0.62, which is lower compared with that of the older population (0.66) reported in Garcia-Molina et al. (2019).

The Bland–Altman graphs for TST and the duration of deep NREM sleep are shown in Figure 5 (a and b, respectively). The former indicates that Smart Sleep (on average) slightly overestimates TST as the bias (mean difference PSG-based TST and Smart Sleep-based TST) is −5 min and the limits-of-agreement (i.e. the interval for 95% of the differences in TST duration) are in the interval −21.6 min to 11.5 min. The Bland–Altman analysis of deep NREM shows that Smart Sleep slightly underestimates the duration of this stage (bias is < 2 min), and 95% of the differences between PSG-based deep NREM sleep and Smart Sleep-based deep NREM sleep are in the interval between −8.1 and 11.9 min.

3.2 | Sleep architecture results

The mean and standard deviation values across study participants for sleep architecture metrics in SHAM and STIM conditions and the results of the Wilcoxon test are presented in Table 3. Based on the selected threshold for statistical significance, none of the sleep architecture metrics is significantly different between SHAM and STIM conditions.

TABLE 2 Sleep staging performance of the Smart Sleep device on a young population.

	Sleep staging			
	Wake	REM	Light NREM	Deep NREM
Sensitivity	0.60	0.67	0.73	0.86
Specificity	0.95	0.95	0.77	0.94

NREM, non-rapid eye movement; REM, rapid eye movement.

The mean and standard deviation values across study participants for stimulation, spectral and SWA properties in SHAM and STIM conditions, along with the results of the Wilcoxon tests are presented on Table 4. Based on the selected threshold for statistical significance, the amplitude of slow waves during stimulation (10.7%), mean SWA (+7.38%) and total SWA (+7.57%) are significantly higher in the STIM condition.

The mean and standard deviation values across study participants for sleepiness and cognitive outcomes in SHAM and STIM conditions and the results of the Wilcoxon test are presented in Table 5. Based on the selected threshold for statistical significance, none of the sleepiness or cognitive metrics is significantly different under SHAM or STIM conditions.

3.3 | Correlation results

The correlations between the STIM/SHAM change in mean and total SWA versus TST, WASO, sleepiness and cognitive outcomes are shown in Table 6. Only three correlations were significant, the negative correlations between the difference in TST between STIM and SHAM, and mean percentage change STIM/SHAM in SWA, mean SWA and total SWA. These correlations are portrayed in Figure 6, where for each participant in the study, the difference in TST between the STIM and SHAM conditions versus the change in mean or total SWA STIM/SHAM are shown.

4 | DISCUSSION

The primary objective of this study was to investigate the feasibility of using closed-loop auditory stimulation to enhance SWA in sleep-restricted adolescents. A device (Smart Sleep) that was proven to increase SWA in adults (Diep et al., 2021; Garcia-Molina et al., 2018; Garcia-molina et al., 2019) was used in this feasibility study.

The ability of Smart Sleep to detect sleep stages in a younger population was evaluated on a subset of participants in this study, each of whom spent a night in the sleep lab simultaneously wearing a PSG system and Smart Sleep. Compared with sleep staging accuracy in adults (Garcia-molina et al., 2019), this test has revealed that, despite a lower overall accuracy (Kappa 0.62 vs. 0.66), the device was able to detect deep NREM sleep with high specificity, which prevents disruptive stimulation, while ensuring better deep NREM detection sensitivity (= 0.86). Consistent with this sensitivity value, the Bland–Altman analysis of deep NREM sleep showed that on average Smart Sleep slightly underestimated the duration of this stage compared with PSG. An important limitation of this evaluation procedure is the manual alignment, based on prominent EEG deflections between the PSG signals and the Smart Sleep data. A better approach to more precisely perform data alignment consists in using bio-calibration events such as ocular movements and blinks.

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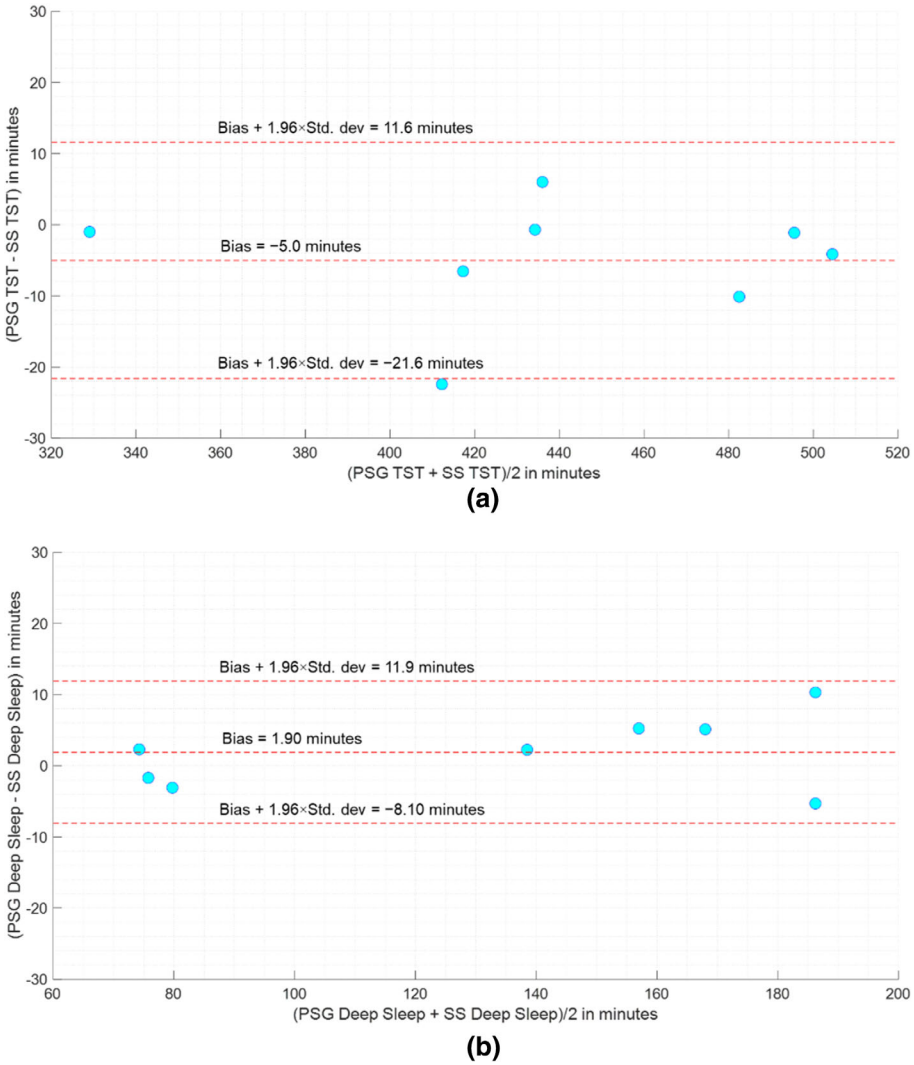

FIGURE 5 Bland–Altman analysis. (a) Total sleep time (TST). (b) Deep non-rapid eye movement (NREM) sleep duration.

TABLE 3 Sleep architecture comparison of SHAM versus STIM.

	SHAM	STIM	*p*-Value
Count of nights	5.0 (1.1)	4.5 (1.3)	0.15
TST (min)	460.0 (35.9)	448.2 (36.6)	0.09
Light NREM duration (min)	230.7 (36.1)	228.3 (40.1)	0.55
Light NREM latency (min)	12.5 (20.0)	7.7 (10.9)	0.09
Deep NREM duration (min)	127.3 (36.6)	129.0 (38.0)	0.94
Deep NREM latency (min)	27.0 (25.0)	24.6 (17.8)	0.91
Deep NREM bout duration (min)	10.7 (3.77)	12.2 (3.91)	0.23
REM duration (min)	102.0 (48.9)	91.0 (37.3)	0.62
REM latency (min)	76.8 (33.4)	69.1 (47.0)	0.43
WASO (min)	29.0 (15.7)	29.7 (21.4)	0.72
Count of interruptions	1.12 (0.88)	1.15 (1.51)	0.72
Count of detected microarousals in NREM	637.3 (651.1)	606.6 (691.4)	0.39
Count of detected microarousals in REM	79.8 (103.1)	84.1 (107.4)	0.73

The table entries represent mean (standard deviation) across study participants.
NREM, non-rapid eye movement; REM, rapid eye movement; SHAM, sham; STIM, stimulation; TST, total sleep time; WASO, wake after sleep onset.

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TABLE 4 Tone count, median volume of the stimulation,[a] spectral and SWA comparison of SHAM versus STIM sessions.

	SHAM	STIM	*p*-Value
Count of tones	2075.6 (886.2)	2248.4 (1225.9)	0.16
Rate of tones (counts per min in deep sleep)	16.7 (4.9)	17.9 (7.5)	0.46
Median volume of the stimulation (dB-SPL)[a]	32.0 (26.0)	32.0 (32.0)	0.55
Count of detected slow waves in deep NREM sleep	5137.6 (1699.9)	5142.5 (1804.6)	0.98
Mean amplitude of detected slow waves in deep NREM sleep (μV)	244.2 (32.6)	246.1 (35.1)	0.89
Count of detected slow waves during stimulation[a]	2901.1 (1342.3)	2935.2 (1415.5)	0.49
Mean amplitude of detected slow waves during stimulation (μV)[a]	277.8 (42.5)	312.3 (99.8)	**0.04***
Mean SWA (μV^2)	1594.7 (556.7)	1715.5 (628.8)	**0.02***
Total SWA ($K\mu V^2$)	5699.8 (2412.4)	6120.5 (2660.3)	**0.01***
Mean alpha power (μV^2)	46.3 (18.9)	47.8 (20.9)	0.33
Mean beta power (μV^2)	13.1 (3.48)	12.9 (3.81)	0.65
Mean sigma power (μV^2)	29.1 (10.4)	29.8 (12.1)	0.52

The table entries represent mean (standard deviation) across study participants.
[a]In the SHAM condition, the periods where the stimulation would have been delivered were considered to estimate volume and properties of detected slow waves.
*Statistically significant values at 0.05 threshold highlighted in bold.
dB-SPL, decibels of sound pressure; NREM, non-rapid eye movement; SHAM, sham; STIM, stimulation; SWA, slow-wave activity.

TABLE 5 Sleepiness and cognitive outcomes comparison of SHAM versus STIM sessions.

	SHAM	STIM	*p*-Value
KSS morning	5.82 (1.12)	6.06 (0.94)	0.25
Mean PVT-RT morning (ms)	286.8 (60.4)	278.4 (48.4)	0.47
# Lapses PVT morning	5.30 (3.46)	5.31 (3.67)	0.88
# False starts PVT morning	3.16 (2.23)	3.12 (2.83)	0.80
N-BACK morning correct responses	54.9 (4.3)	56.0 (3.8)	0.07
KSS evening	5.63 (1.5)	5.53 (1.8)	0.42
Mean PVT-RT evening (ms)	278.8 (47.4)	268.8 (30.1)	0.33
# Lapses PVT evening	4.62 (3.63)	4.23 (2.46)	0.61
# False starts PVT evening	2.44 (1.73)	2.17 (1.53)	0.76
N-BACK evening correct responses	55.9 (3.8)	56.1 (3.0)	0.61

The table entries represent mean (standard deviation) across study participants.
KSS, Karolinska Sleepiness Scale; N-BACK, continuous working memory performance task; PVT, Psychomotor Vigilance Task; RT, reaction time; SHAM, sham; STIM, stimulation.

TABLE 6 Pearson's correlation between mean (during stimulation) SWA, SWA change STIM/SHAM, and sleep architecture, sleepiness and cognitive outcome values STIM-SHAM.

	Mean % change in (during stimulation) SWA STIM/SHAM	Mean SWA % change STIM/SHAM	Mean total SWA % change STIM/SHAM
TST STIM–SHAM (min)	**−0.74** ($p < 10^{-3}$)	**−0.63** ($p = 7.1 \times 10^{-3}$)	**−0.83** ($p < 10^{-4}$)
WASO STIM—SHAM (min)	0.06 ($p = 0.81$)	0.38 ($p = 0.13$)	0.18 ($p = 0.18$)
KSS morning STIM–SHAM	0.45 ($p = 0.31$)	0.35 ($p = 0.44$)	0.63 ($p = 0.13$)
Mean PVT-RT morning STIM–SHAM (ms)	0.25 ($p = 0.35$)	0.06 ($p = 0.82$)	0.07 ($p = 0.75$)
Mean N-BACK morning STIM–SHAM correct responses	0.23 ($p = 0.28$)	0.11 ($p = 0.69$)	0.02 ($p = 0.95$)
KSS evening STIM–SHAM	0.11 ($p = 0.69$)	−0.04 ($p = 0.88$)	0.06 ($p = 0.82$)
Mean PVT-RT evening STIM–SHAM (ms)	−0.02 ($p = 0.95$)	−0.19 ($p = 0.49$)	−0.27 ($p = 0.32$)
Mean N-BACK evening STIM–SHAM correct responses	−0.20 ($p = 0.48$)	−0.28 ($p = 0.31$)	−0.24 ($p = 0.38$)

Note: Statistically significant values highlighted in bold.
KSS, Karolinska Sleepiness Scale; N-BACK, continuous working memory performance task; PVT, Psychomotor Vigilance Task; RT, reaction time; SHAM, sham; STIM, stimulation; SWA, slow-wave activity; TST, total sleep time; WASO, wake after sleep onset.

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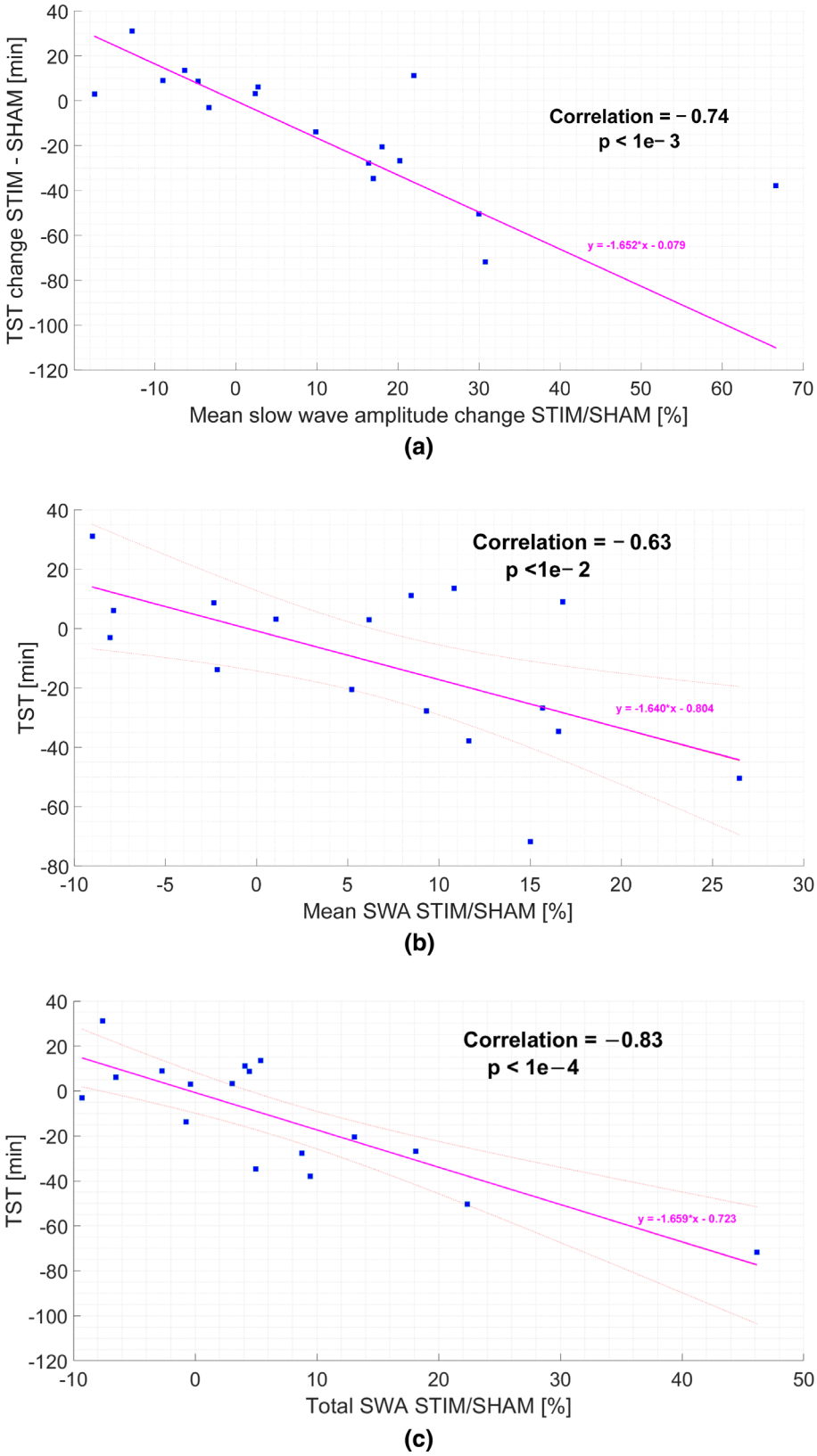

FIGURE 6 STIM–SHAM difference in total sleep time (TST) versus: (a) mean slow-wave amplitude (SWA) change of slow waves during stimulation STIM/SHAM; (b) mean SWA change STIM/SHAM; and (c) total SWA change STIM/SHAM.

We found a significant increase in both mean and total SWA during nights with auditory stimulation compared with SHAM (+7.38%, $p = 0.01$ and + 7.57%, $p = 0.02$, respectively). The amplitude of detected slow waves during stimulation was also significantly higher in the STIM condition (+10.7%, $p = 0.04$). The count of slow waves and the mean amplitude of slow waves across the entire period of deep NREM sleep were not significantly different. This suggests that the effect of stimulation manifests as a SWA increase limited to when the stimulation is present.

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To assess a potential cumulative effect on SWA of the stimulation (across consecutive nights), we compared the mean SWA of the last night in the STIM condition versus the last night in the SHAM condition. On average, the mean SWA enhancement across all participants in the study was 17.8%. This is an interesting result that motivates a dedicated study specifically aimed at quantifying cumulative effects of consecutive STIM that may need to be conducted in-lab to ensure protocol adherence.

Despite the increase in SWA being evident, our analysis did not reveal significant alterations in overall sleep architecture between stimulation and sham conditions. However, we did observe a significant negative correlation between the difference in TST and the change in mean and total SWA between STIM and SHAM conditions, suggesting a potential relationship between sleep duration and SWA enhancement. Within the framework of the S process (homeostatic) component of the two-process model of sleep regulation, SWA is associated with sleep-need dissipation (Achermann & Borbély, 1990, 2003; Borbély, 1982). Thus, an increase in SWA may optimize sleep-need dissipation, which could imply a shorter sleep duration. Such an effect may be more pronounced in sleep-deprived individuals such as the ones who participated in this research.

The spectral comparison in the alpha, sigma and beta power did not show any statistically significant differences between SHAM and STIM. The absence of significant differences in alpha and beta bands is consistent with the analysis of microarousals and suggests that the stimulation did not disturb sleep. The power in the sigma band reflects spindle activity, which has been reported to be enhanced by auditory stimulation (Ngo et al., 2013). Spindle activity is better observed at central EEG sites (Adra et al., 2022), which may explain the absence of significant differences in SHAM versus STIM conditions as characterized by Smart Sleep, which uses a frontal derivation.

Exploring the secondary objective of assessing the impact of SWA enhancement on cognitive function, we examined performance metrics derived from PVT and N-BACK. Although no significant differences were observed in cognitive outcomes between stimulation and sham conditions, our correlation analysis revealed intriguing associations between changes in SWA and certain cognitive measures. Trending results ($p = 0.07$) in the STIM condition, presumably due to the limited sample, included an improvement in the number of correct morning N-BACK responses (+1.01 correct).

Several limitations of this feasibility trial should be acknowledged. Firstly, the small sample size limits the generalizability of the findings, emphasizing the need for larger-scale studies to validate the efficacy of closed-loop auditory stimulation in enhancing SWA and improving sleep quality in adolescents. Additionally, the short duration of the intervention period may not fully capture the long-term effects of auditory stimulation on sleep patterns and cognitive function. Given that auditory stimulation, by virtue of increasing SWA, may modify the sleep-need dissipation as per the two-process model, cumulative effects across subsequent days of stimulation may exist. These need to be analysed by designing a protocol with well-controlled washout periods between SHAM and STIM conditions, and a strict number of sleep sessions under each condition. Finally, the algorithm embedded in the Smart Sleep device was trained to recognize slow waves in a population of adults, and therefore it may not be optimized to deliver stimulation in a younger population. Although by late adolescence the morphology and distribution of slow waves resembles that of adults, with the highest amplitude of slow waves concentrated in frontal brain regions, slow-wave morphology and distribution may not have reached maturity in the younger adolescents in our sample (Kurth et al., 2010; Timofeev et al., 2020).

Future research activities should explore the sustained impact of this intervention over an extended duration and consider individual variability in response to auditory stimuli. Because the original algorithm to detect sleep stages was trained with the data from adults in the 20–45-year-old range, retraining the algorithm with data from a younger cohort is warranted. Improvement aspects include: (1) retraining the DNN-based sleep staging, possibly leveraging a transfer learning strategy, with data from adolescent subjects; and (2) adjusting the stimulation delivery parameters by adjusting the sleep-depth thresholds (d_m and d_M) using the data from those subjects. This can also enable the possibility of delivering stimulation in light NREM sleep.

5 | CONCLUSION

Our feasibility trial provides preliminary evidence supporting the effectiveness of closed-loop auditory stimulation in enhancing SWA during sleep in sleep-restricted adolescents. While further research is warranted to elucidate the broader implications of this intervention on sleep architecture and cognitive function, these findings contribute to the growing body of literature exploring innovative strategies to address adolescent sleep deprivation and its associated health consequences. By harnessing technological advancements in sleep science, such as closed-loop auditory stimulation, we endeavour to pave the way for personalized interventions aimed at optimizing sleep health and promoting overall well-being in adolescents.

AUTHOR CONTRIBUTIONS

Gary Garcia Molina: Software; formal analysis; writing – review and editing; visualization; writing – original draft; investigation. **Camilla Matthews:** Conceptualization; writing – original draft; investigation. **Annika Myers:** Data curation; resources; methodology. **Bethany Flaherty:** Investigation; methodology; validation; resources. **Emma Strainis:** Writing – review and editing. **Brady Riedner:** Conceptualization; software; writing – review and editing; writing – original draft. **Ana Maria Vascan:** Methodology; resources; data curation. **Giulio Tononi:** Supervision; funding acquisition; writing – original draft; writing – review and editing; methodology. **Stephanie Jones:** Conceptualization; investigation; funding acquisition; writing – original draft; methodology; validation; writing – review and editing; project administration; supervision; resources.

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CONFLICT OF INTEREST STATEMENT

G.G. was affiliated with Philips Sleep and Respiratory Care from August 2017 to May 2020. He led some aspects in the development of the Smart Sleep Deep Sleep Headband, which was used in the research reported in this article.

DATA AVAILABILITY STATEMENT

The data presented in this article can be made available upon request. Requests to access the datasets should be directed to the corresponding author.

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REFERENCES

Achermann, P., & Borbély, A. A. (1990). Simulation of human sleep: Ultradian dynamics of electroencephalographic slow-wave activity. *Journal of Biological Rhythms*, *5*(2), 141–157. https://doi.org/10.1177/074873049000500206

Achermann, P., & Borbély, A. A. (2003). Mathematical models of sleep regulation. *Frontiers in Bioscience: a Journal and Virtual Library*, *8*(13), s683–s693. http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=PubMed&dopt=Citation&list_uids=12700054

Adra, N., Sun, H., Ganglberger, W., Ye, E. M., Dümmer, L. W., Tesh, R. A., Westmeijer, M., Cardoso, M. D. S., Kitchener, E., Ouyang, A., Salinas, J., Rosand, J., Cash, S. S., Thomas, R. J., & Westover, M. B. (2022). Optimal spindle detection parameters for predicting cognitive performance. *Sleep*, *45*(4), zsac001. https://doi.org/10.1093/sleep/zsac001

Akerstedt, T., Hume, K., Minors, D., & Waterhouse, J. (1997). Good sleep – its timing and physiological sleep characteristics. *Journal of Sleep Research*, *6*(4), 221–229.

Basner, M., Mollicone, D., & Dinges, D. F. (2011). Validity and sensitivity of a brief psychomotor vigilance test (PVT-B) to Total and partial sleep deprivation. *Acta Astronautica*, *69*(11–12), 949–959. https://doi.org/10.1016/j.actaastro.2011.07.015

Bellesi, M., Riedner, B., Garcia-Molina, G., Cirelli, C., & Tononi, G. (2014). Enhancement of sleep slow waves: Underlying mechanisms and practical consequences. *Frontiers in Systems Neuroscience*, *8*(October), 1–17. https://doi.org/10.3389/fnsys.2014.00208

Berry, R. B., Brooks, R., Gamaldo, C. E., Harding, S. M., Lloyd, R. M., Marcus, C. L., Vaughn, B. V., & Thomas, S. M. (2015). *The AASM manual for the scoring of sleep and associated events: Rules terminology and technical specifications, version 2.0*. American Academy of Sleep Medicine.

Bland, J. M., & Altman, D. G. (1986). Statistical methods for assessing agreement between two methods of clinical measurement. *Lancet*, *1*, 307–310.

Borbély, A. A. (1982). A two process model of sleep regulation. *Human Neurobiology*, *1*(3), 195–204.

Bresch, E., Grossekathofer, U., & Garcia-Molina, G. (2018). Recurrent deep neural networks for real-time sleep stage classification from single channel EEG. *Frontiers in Computational Neuroscience*, *12*(2018), 1–12. https://doi.org/10.3389/fncom.2018.00085

Carskadon, M. A. (2011). Sleep in adolescents: The perfect storm. *Pediatric Clinics of North America*, *58*(3), 637–647. https://doi.org/10.1016/j.pcl.2011.03.003

Cohen, J., Cohen, P., West, S. G., & Aiken, L. S. (2003). *Applied multiple regression/correlation analysis for the behavioral sciences* (3rd ed.). Lawrence Erlbaum Associates.

Diep, C., Garcia-Molina, G., Jasko, J. G., Manousakis, J. E., Ostrowski, L., White, D. P., & Anderson, C. (2021). Acoustic enhancement of slow wave sleep on consecutive nights improves alertness and attention in chronically short sleepers. *Sleep Medicine*, *81*, 69–79. https://doi.org/10.1016/j.sleep.2021.01.044

Dijk, D.-J. (2009). Regulation and functional correlates of slow wave sleep. *Journal of Clinical Sleep Medicine*, *5*(2 Suppl), S6–S15.

Dinges, D. F., & Powell, J. W. (1985). Microcomputer analyses of performance on a portable, simple visual RT task during sustained operations. *Behavior Research Methods, Instruments, & Computers*, *17*(6), 652–655.

Garcia-Molina, G., Tsoneva, T., Jasko, J., Steele, B., Aquino, A., Baehr, K., Pastoor, S., Pfundtner, S., Ostrowski, L., Miller, B., Papas, N., Riedner, B., Tononi, G., & White, D. P. (2018). Closed-loop system to enhance slow-wave activity. *Journal of Neural Engineering*, *15*(6), 1–11. https://doi.org/10.1088/1741-2552/aae18f

Garcia-molina, G., Tsoneva, T., Neff, A., Salazar, J., Bresch, E., Grossekathofer, U., Pastoor, S., & Aquino, A. (2019). Hybrid in-phase and continuous auditory stimulation significantly enhances slow wave activity during sleep. In *2019 41st annual international conference of the IEEE engineering in medicine and biology society (EMBC)* (pp. 4052–4055). IEEE.

Garcia-Molina, G., Vissapragada, S., Mahadevan, A., Goodpaster, R., Riedner, B., Bellesi, M., & Tononi, G. (2016). Probabilistic characterization of sleep architecture: Home based study on healthy volunteers. In *2016 38th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)* (pp. 2834–2838). IEEE.

Gradisar, M., Kahn, M., Micic, G., Short, M., Reynolds, C., Orchard, F., Bauducco, S., Bartel, K., & Richardson, C. (2022). Sleep's role in the development and resolution of adolescent depression. *Nature Reviews Psychology*, *1*(9), 512–523. https://doi.org/10.1038/s44159-022-00074-8

Huber, R., Felice Ghilardi, M., Massimini, M., & Tononi, G. (2004). Local sleep and learning. *Nature*, *430*(6995), 78–81. https://doi.org/10.1038/nature02663

Jamieson, D., Shan, Z., Lagopoulos, J., & Hermens, D. F. (2021). The role of adolescent sleep quality in the development of anxiety disorders: A neurobiologically-informed model. *Sleep Medicine Reviews*, *59*, 101450. https://doi.org/10.1016/j.smrv.2021.101450

Kaida, K., Takahashi, M., Akerstedt, T., Nakata, A., Otsuka, Y., Haratani, T., & Fukasawa, K. (2006). Validation of the Karolinska sleepiness scale against performance and EEG variables. *Clinical Neurophysiology*, *117*(7), 1574–1581. https://doi.org/10.1016/j.clinph.2006.03.011

Kirchner, W. K. (1958). Age differences in short-term retention of rapidly changing information. *Journal of Experimental Psychology*, *55*(4), 352–358.

Kurth, S., Ringli, M., Geiger, A., LeBourgeois, M., Jenni, O. G., & Huber, R. (2010). Mapping of cortical activity in the first two decades of life: A high-density sleep electroencephalogram study. *The Journal of Neuroscience*, *30*(40), 13211–13219. https://doi.org/10.1523/JNEUROSCI.2532-10.2010

Leminen, M. M., Virkkala, J., Saure, E., Paajanen, T., Zee, P. C., Santostasi, G., Hublin, C., Müller, K., Porkka-Heiskanen, T., Huotilainen, M., & Paunio, T. (2017). Enhanced memory consolidation via automatic sound stimulation during non-REM sleep. *Sleep*, *40*(3), zsx003. https://doi.org/10.1093/sleep/zsx003

Lo, J. C., & Chee, M. W. (2020). Cognitive effects of multi-night adolescent sleep restriction: Current data and future possibilities. *Current Opinion in Behavioral Sciences*, *33*, 34–41. https://doi.org/10.1016/j.cobeha.2019.12.005

Lo, J. C., Ong, J. L., Leong, R. L. F., Gooley, J. J., & Chee, M. W. L. (2016). Cognitive performance, sleepiness, and mood in partially sleep

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deprived adolescents: The need for sleep study. *Sleep*, *39*(3), 687–698. https://doi.org/10.5665/sleep.5552

Ngo, H. V., Martinetz, T., Born, J., & Mölle, M. (2013). Auditory closed-loop stimulation of the sleep slow oscillation enhances memory. *Neuron*, *78*(3), 545–553. https://doi.org/10.1016/j.neuron.2013.03.006

Owens, J. A., & Weiss, M. R. (2017). Insufficient sleep in adolescents: causes and consequences. *Minerva Pediatrics*, *69*(4), 326–336. https://doi.org/10.23736/S0026-4946.17.04914-3

Palmer, C. A., & Alfano, C. A. (2017). Sleep and emotion regulation: An organizing, integrative review. *Sleep Medicine Reviews*, *31*, 6–16. https://doi.org/10.1016/j.smrv.2015.12.006

Papalambros, N. A., Santostasi, G., Malkani, R. G., Braun, R., Weintraub, S., Paller, K. A., & Zee, P. C. (2017). Acoustic enhancement of sleep slow oscillations and concomitant memory improvement in older adults. *Frontiers in Human Neuroscience*, *11*(March), 1–14. https://doi.org/10.3389/fnhum.2017.00109

Paruthi, S., Brooks, L. J., D'Ambrosio, C., Hall, W. A., Kotagal, S., Lloyd, R. M., Malow, B. A., Maski, K., Nichols, C., Quan, S. F., Rosen, C. L., Troester, M. M., & Wise, M. S. (2016). Consensus Statement of the American Academy of sleep medicine on the recommended amount of sleep for healthy children: Methodology and discussion. *Journal of Clinical Sleep Medicine*, *12*(11), 1549–1561. https://doi.org/10.5664/jcsm.6288

Rey, D., & Neuhäuser, M. (2011). Wilcoxon-signed-rank test. In M. Lovric (Ed.), *International encyclopedia of statistical science*. Springer, Berlin.

Short, M. A., & Chee, M. W. L. (2019). Adolescent sleep restriction effects on cognition and mood. *Progress in Brain Research*, *246*, 55–71. https://doi.org/10.1016/bs.pbr.2019.02.008

Spilsbury, J. C., Drotar, D., Rosen, C. L., & Redline, S. (2007). The Cleveland adolescent sleepiness questionnaire. *Journal of Clinical Sleep Medicine*, *3*(6), 603–612.

Timofeev, I., Schoch, S. F., LeBourgeois, M. K., Huber, R., Riedner, B. A., & Kurth, S. (2020). Spatio-temporal properties of sleep slow waves and implications for development. *Current Opinion in Physiology*, *15*, 172–182. https://doi.org/10.1016/j.cophys.2020.01.007

Tononi, G., & Cirelli, C. (2006). Sleep function and synaptic homeostasis. *Sleep Medicine Reviews*, *10*(1), 49–62. https://doi.org/10.1016/j.smrv.2005.05.002

Tononi, G., & Cirelli, C. (2014). Sleep and the price of plasticity: From synaptic and cellular homeostasis to memory consolidation and integration. *Neuron*, *81*(1), 12–34. https://doi.org/10.1016/j.neuron.2013.12.025

Uccella, S., Cordani, R., Salfi, F., Gorgoni, M., Scarpelli, S., Gemignani, A., Geoffroy, P. A., De Gennaro, L., Palagini, L., Ferrara, M., & Nobili, L. (2023). Sleep deprivation and insomnia in adolescence: Implications for mental health. *Brain Sciences*, *13*(4), 569. https://doi.org/10.3390/brainsci13040569

Watson, N., Badr, M. S., Belenky, G., Bliwise, D. L., Buxton, O. M., Buysse, D., Dinges, D. F., Gansgwich, J., Grandner, M. A., Kushida, C., Malhotra, R. K., Martin, J. L., Patel, S. R., Quan, S. F., & Tasali, E. (2015). Recommended amount of sleep for a healthy adult : A joint consensus Statement of the American Academy of sleep medicine and Sleep Research Society. *Sleep*, *38*(6), 843–844.

Welch, P. (1967). The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms. *IEEE Transactions on Audio and Electroacoustics*, *15*(2), 70–73. https://doi.org/10.1109/TAU.1967.1161901

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